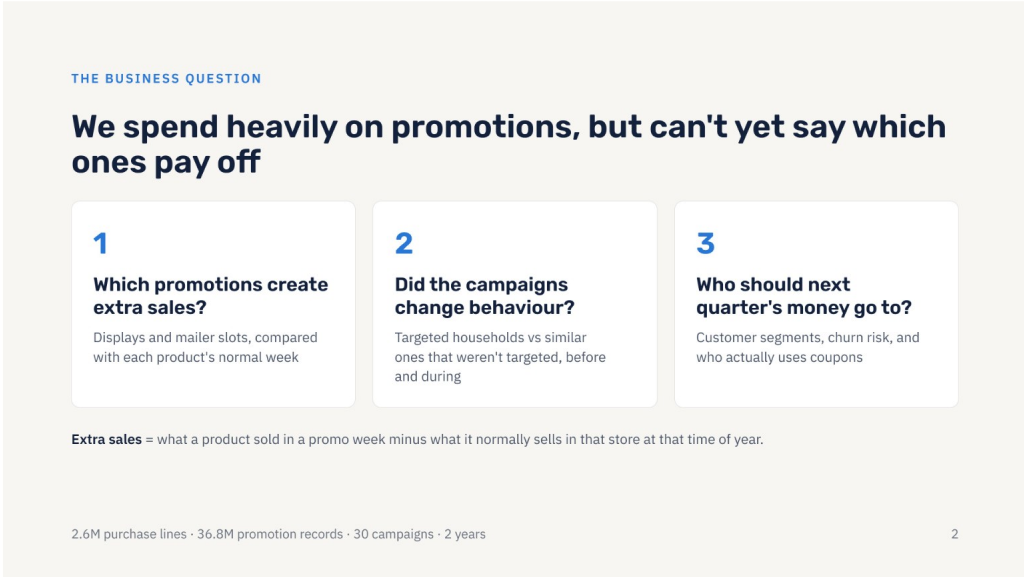
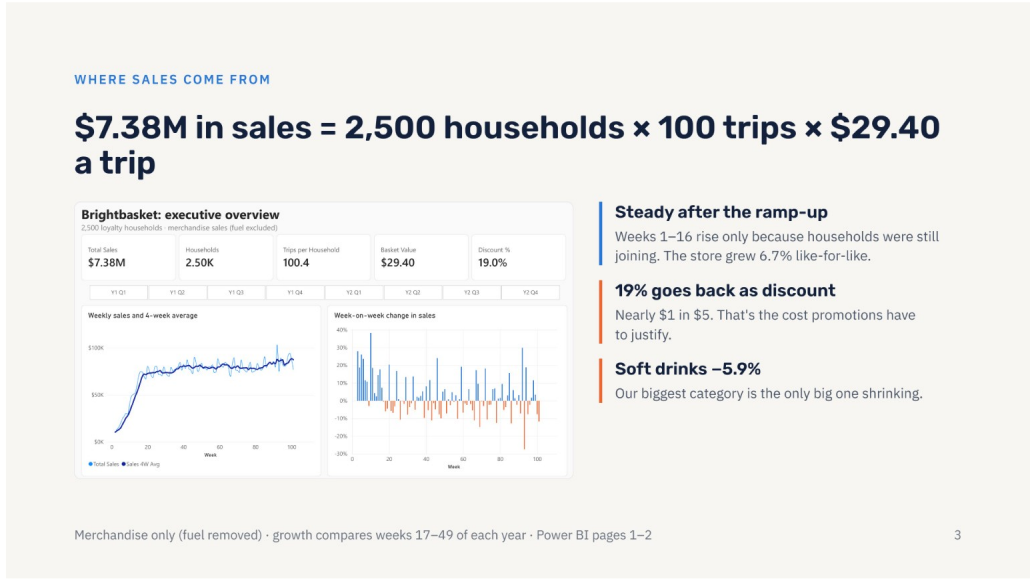


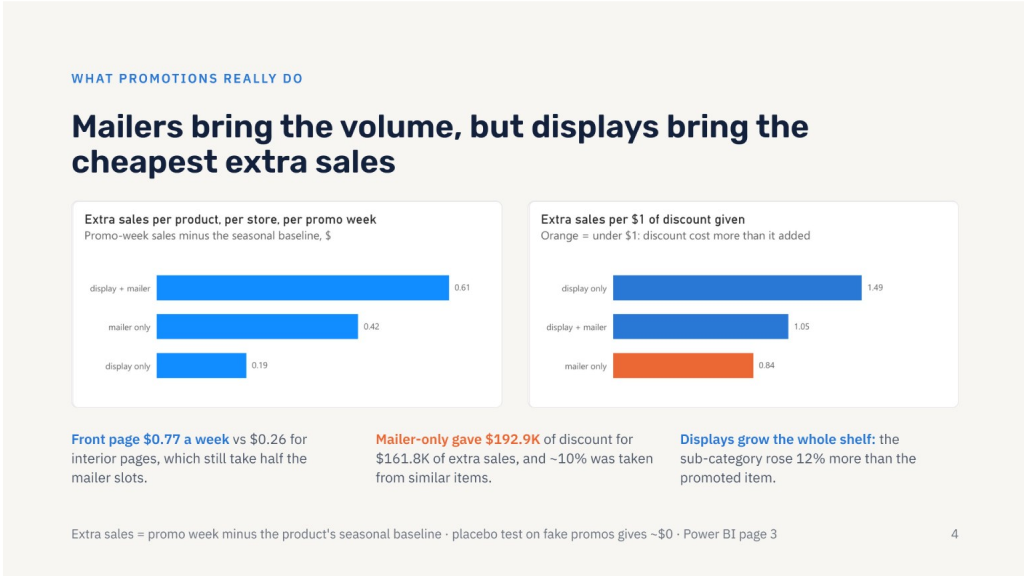
Thanks for the time. Brightbasket spends a lot on promotions: in-store displays, the weekly mailer, targeted campaigns and coupons. The question I set out to answer is simple: which of these actually bring in sales we would not have made anyway, and where should next quarter's money go. I'll go through what the data shows, what it can't show, and end with four concrete moves.



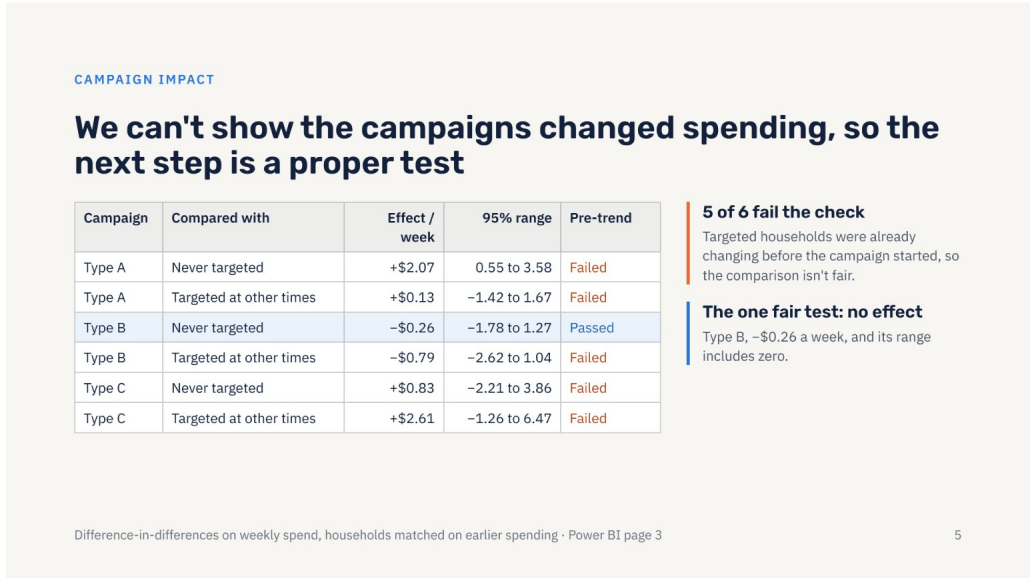
Three questions drive everything that follows. The key idea is "extra sales": not what sold during a promotion, but what sold on top of a normal week. A lot of promoted sales would have happened anyway, and those are just discounts we gave away.



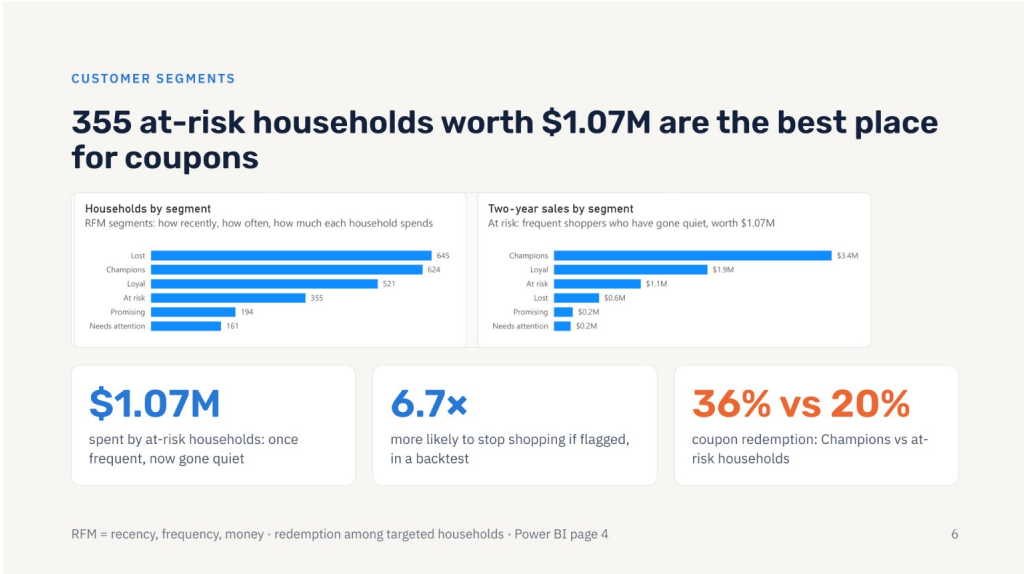
This is the sales tree. Every change in sales has to come from more households, more trips, or bigger baskets. The early climb is the panel filling up, not growth; comparing the same weeks of each year, the store grew 6.7%. The number to hold onto is 19%: that's how much of sales value goes back to customers as discounts. And one category flag: soft drinks is the biggest category and it's shrinking, which matters because it's also heavily promoted.



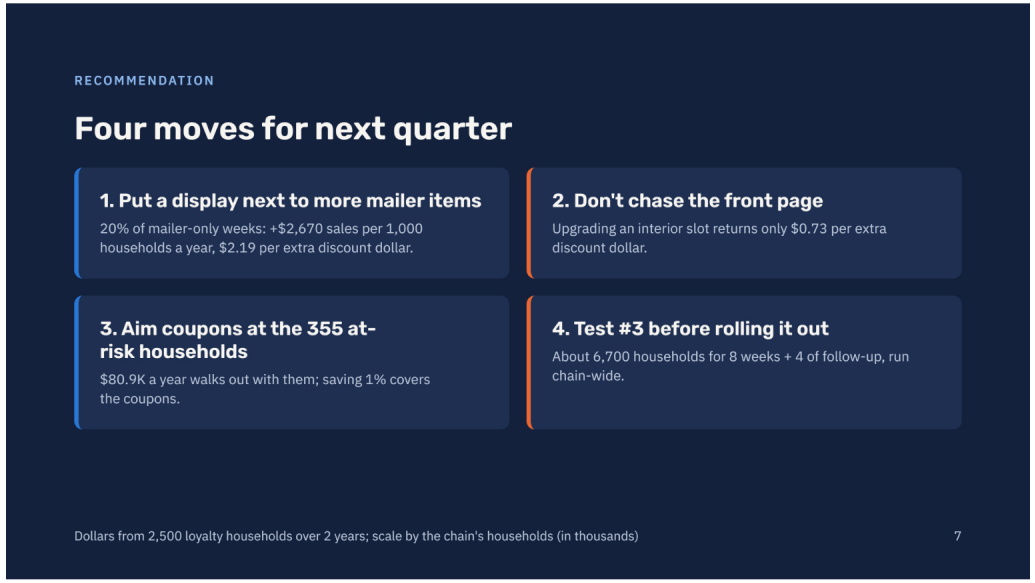
For every promo week I compared the product's sales with its normal week in the same store, adjusted for season. On the left is volume: putting a product in the mailer does most of the work, and the front page earns three times what interior pages do. On the right is cost: per dollar of discount, displays are the best deal we have, about a dollar fifty, while mailer-only weeks come in under a dollar. Mailers also borrow around a tenth of their gain from similar products, whereas displays lift the whole shelf. To check the method wasn't inventing uplift, I ran it on fake promo weeks twenty times and got basically zero. One caveat: this compares discount with sales, not margin.



For campaigns I compared how weekly spend changed for targeted households against similar households that weren't targeted, before versus during. That only works if both groups were moving in step beforehand, and for five of the six comparisons they weren't. My read is that campaigns were sent to households whose spending was already shifting, a kind of win-back targeting, so I can't separate the campaign from what would have happened anyway. The one comparison that passes shows nothing. That's not proof campaigns don't work; it's proof we haven't measured it yet. Which is exactly why the last recommendation is a test.



I scored every household on how recently, how often and how much they shop, and grouped them. Champions spend the most, $3.4M, and they redeem coupons the most, which is the problem: they'd mostly buy anyway, so a coupon there is a discount, not a sale. The at-risk group is different. These were frequent shoppers who have gone quiet, they're worth a million dollars, and only one in five has used a coupon. I also built a simple churn flag and tested it on an earlier cut of the data: households it flagged were 6.7 times more likely to actually leave. Higher-income households redeem more too, about 52% against 35% for the lowest band.



Four moves. First, give more mailer items a display: display and mailer simply add up, so it's the cheapest extra sales we can buy, about $2.19 for every extra discount dollar. It needs about a fifth more display space, which is the real constraint. Second, a "don't": moving interior-page features to the front page looks tempting because the front page sells three times more, but it's discounted so much harder that it returns less than a dollar per discount dollar. Third, coupons: the at-risk group is expected to take about $81K a year with them, and the coupons we now give Champions cost under $1K, so saving even 1% of those households pays for itself. But I haven't proven the offer saves anyone, which is the fourth move: a randomised test. To detect a 10% lift in weekly spend I'd need about 6,700 households, more than this sample has, so it runs across the chain.